

Process Redesign

Lecture 07

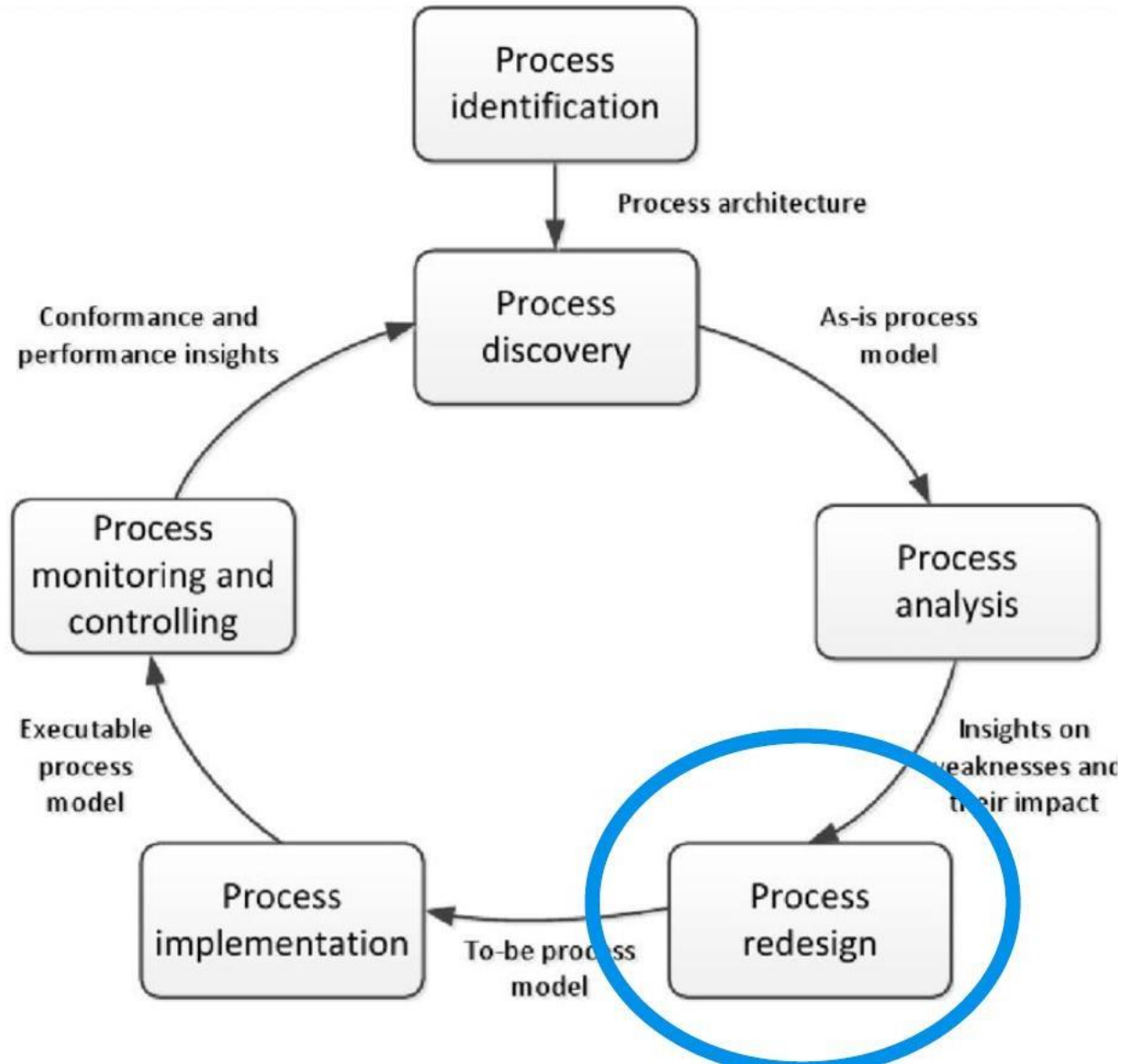
BUSINESS PROCESS MANAGEMENT
IS 2206

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OUTLINE

1. Introduction
 1. 7 Elements to decide on Process Redesign
 2. What is Process Redesign
 3. Types of Organizations
 4. Levels of Abstraction in Process Redesign
 5. Approaches of Process Redesign
2. Methodologies of Process Redesign
 1. Heuristic Process Redesign
 2. Product Based Design





TOPIC 01

What is Process Discovery



1.1 What is Process Redesign?

It is all about improving the quality of products and services by rethinking and re-organizing business processes.



1.2 Why Process Redesign

- Adapting to Change
- Improving Efficiency
- Addressing Performance Issues
- Enabling Innovation

1.3 Seven(7) Elements to decide on Process Redesign

1. Internal / External Customers
2. Business Process Operation (No. of activities, nature)
3. Business Process Behaviour(Order / Schedule of activities)
4. Organization
5. Information
6. Technology
7. External Environment

1.4 Redesign?

1. **First** : changing the business process itself, covering both its operational and behavioral view
2. **Second**: Organization, External Environment, Information, Technology
3. **Third**: Products it delivers to customers

1.5 Organization Types

- **Manufacturing Organizations**

- Transforming raw materials into tangible products, which often relies on the use of robots and sophisticated machinery.

- **Service Organizations**

- Knowledge is involved in the processing of information to deliver a particular service.



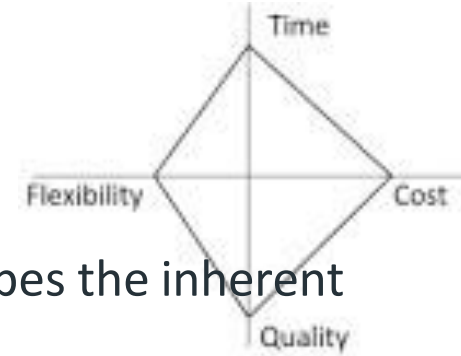
Manufacturing Organizations

- Making a copy is relatively difficult
- Requires space for inventory
- Needs a proper order for the machinery activities
- Relatively difficult to measure quality
- Transportation of inventory is time consuming

Service Organizations

- Making a copy is relatively easier
- Does not require much space
- Less requirements for the order of the activities (Humans are more flexible)
- Relatively easier to measure quality
- Transportation happens through networks

1.6 Directions For Improvement



The Devil's Quadrangle

- The Devil's Quadrangle is a framework in Business Process Management (BPM) that describes the inherent trade-offs between four key performance dimensions: time, cost, quality, and flexibility.
- It highlights that improving one of these dimensions often comes at the expense of another, making it a "devilish" challenge to optimize all aspects of a process simultaneously.
 - **Time:** Refers to the duration of a process, with the goal of minimizing it for faster turnaround times.
 - **Cost:** Encompasses all expenses associated with the process, aiming to reduce operational costs.
 - **Quality:** Focuses on the accuracy, reliability, and effectiveness of the process, with the goal of maximizing it.
 - **Flexibility:** Relates to the adaptability of the process to changing circumstances and requirements, aiming for adaptability and responsiveness.
- The Devil's Quadrangle is used to analyze and understand these trade-offs, helping organizations make informed decisions about process design and improvement. For instance, streamlining a process to reduce time might lead to increased costs or decreased quality. Similarly, enhancing quality could require more time and resources, impacting cost and flexibility.

1.7 Approaches to Process Redesign

1. **Clean Slate Approach:** Start from scratch omitting existing processes
2. **Improve Existing Approach:** From the traits of an existing process
3. **Blueprint / Reference Model:** From a good general design



TOPIC 02

Methodologies of Process Redesign



2. What?

- The two methodologies for process redesign are
 1. Heuristic Process Redesign and
 2. Product-Based Design.
- Heuristic Process Redesign focuses on using established rules and guidelines to improve processes, while Product-Based Design centers on creating new processes aligned with a specific product or service.
- In essence, Heuristic Process Redesign focuses on refining existing processes, while Product-Based Design concentrates on creating new processes from the ground up, tailored to a specific product or service

2.1 Heuristic Process Redesign - Approach

1. Initiate

- Setup Redesign Project
- Setup the team
- Understanding the As – Is model
- Setting goals for redesign output

2. Design

- Use the fixed list of redesign heuristics
- Decide improvements on each heuristic

3. Evaluate

- Evaluating qualitatively or quantitatively
- Using simulations to evaluate

2.1 Heuristic Process Redesign - Approach

**7 Elements to
decide on
Process
Redesign**

=

**7 Redesign
Heuristics**

2.1.1 Customer Heuristics

Focusses on improving the interaction with customers by;

- **Control Relocation** –Move the control towards customer
- **Contact Reduction** -Reduce the number of contacts with customers and third parties
- **Integration**-Consider the integration with a business process of the customer or a supplier

Effect of Customer Heuristics on Devil's Quadrangle

	Time	Cost	Quality	Flexibility
Control relocation	.	—	+	.
Contact reduction	+	—	+	.
Integration	+	+	.	—

+ Positive
- Negative
. Neutral

2.1.2 Business Process Operation Heuristics

Focusses on the elements of a business process.

- **Case Types** - Determine whether activities are related to the same type of case and, if necessary, distinguish new business processes
- **Activity Elimination** - Eliminate unnecessary activities from a business process.
- **Case Based Work** - Consider removing batch-processing and periodic activities from a business process
- **Triage** - Consider the division of a general activity into two or more alternative activities
- **Activity Composition** - Combine small activities into composite activities and divide large activities into workable smaller activities

Effect of Business Process Operation Heuristics on Devil's Quadrangle

	Time	Cost	Quality	Flexibility
Case types	+	+	—	—
Activity elimination	+	+	—	.
Case-based work	+	—	.	.
Triage	.	—	+	—
Activity composition	+	+	.	—

+ Positive
- Negative
. Neutral



2.1.3 Business Process Behavior Heuristics

Focusses on regulating the logic within the business process

- **Resequencing** – Move activities to more appropriate places
- **Parallelism** – Consider whether activities may be executed in parallel
- **Knock –Out** – Order knock-outs in an increasing order of effort and in a decreasing order of termination probability
- **Exception** - Design business processes for typical cases and isolate exceptional cases from the normal flow

Effect of Business Process Behaviour Heuristics on Devil's Quadrangle

	Time	Cost	Quality	Flexibility
Resequencing	+	+	.	.
Parallelism	+	–	.	–
Knock-out	–	+	.	.
Exception	+	–	+	–

+ Positive
- Negative
. Neutral



2.1.4 Organization Heuristics

There are two sets of Organization Heuristics.

- First set focusses on the structure of the organization (mostly the allocation of resources).
- There are 07 heuristics in this set
 - **Case Assignment** - Let workers perform as many steps as possible for single cases
 - **Flexible Assignment** - Assign work in such a way that maximal flexibility is preserved for the near future
 - **Centralization** - Treat geographically dispersed resources as if they are centralized
 - **Split Responsibilities** - Avoid shared responsibilities for tasks by people from different functional units
 - **Customer Teams** - Consider to compose work teams of people from different departments that will take care of the complete handling of specific sorts of cases.
 - **Numerical Involvement** - Minimize the number of departments, groups and persons involved in a business process
 - **Case Manager** - Appoint one person to be responsible for the handling of each type of case, the case manager

Effect of Organization Heuristics on Devil’s Quadrangle (First Set)

	Time	Cost	Quality	Flexibility
Case assignment	.	.	+	—
Flexible assignment	+	—	.	+
Centralization	+	—	.	+
Split responsibilities	.	.	+	—
Customer teams	.	.	+	—
Numerical involvement	+	—	.	—
Case manager	.	—	+	.

+ Positive
 - Negative
 . Neutral

2.1.4 Organization Heuristics (Contd)

There are two sets of Organization Heuristics.

- Second set focusses on the organizational population and the resources being involved in terms of type and number.
- There are 03 heuristics in this set
 - **Extra Resources** If capacity is insufficient, consider increasing the available number of resources
 - **Specialist Generated** Consider to deepen or broaden the skills of resources
 - **Empower** - Give workers most of the decision making authority instead of relying on middle management

Effect of Organization Heuristics on Devil's Quadrangle (Second Set)

	Time	Cost	Quality	Flexibility
Extra resources	+	—	.	+
Specialist-generalist	+	.	+	—
Empower	+	.	—	+

+ Positive
- Negative
. Neutral



2.1.5 Information Heuristics

Focusses on the information the business process uses, creates, may use or may create

- **Control Addition** – Check the completeness and correctness of incoming materials and check the output before it is sent to customers
- **Buffering** – Instead of requesting information from an external source, buffer it and subscribe to updates.

Effect of Information Heuristics on Devil's Quadrangle

	Time	Cost	Quality	Flexibility
Control addition	–	–	+	.
Buffering	+	–	.	.

+ Positive
- Negative
. Neutral



2.1.6 Technology Heuristics

Focusses on the technology the business process uses or may use.

○ **Activity Automation** – Consider automating activities

○ **Integral Technology** – Try to elevate physical constraints in a business process by applying new technology

Effect of Technology Heuristics on Devil's Quadrangle

	Cost	Quality	Time	Flexibility
Activity automation	+	–	+	–
Integral technology	+	–	.	.



+ Positive
– Negative
· Neutral

2.1.7 External Environment Heuristics

Focusses on the heuristics that try to improve upon the collaboration and communication with the third parties

- **Trusted Party** - Instead of determining information oneself, use the results of a trusted party
- **Outsourcing** - Consider outsourcing a business process completely or parts of it.
- **Interfacing** - Consider a standardized interface with customers and partners

Effect of External Environment Heuristics on Devil's Quadrangle

	Cost	Quality	Time	Flexibility
Trusted party	+	+	.	-
Outsourcing	+	+	.	-
Interfacing	+	.	+	-



+ Positive
- Negative
. Neutral

2.2.1 Product Based Design – What?

- Aims at radically rethinking how a particular product or service can be created instead of using an incremental approach
- Existing process is not always the starting point of redesign. Rather, the characteristics of the particular product that the process-to-be is expected to deliver are used to reason back what that process should look like.
- By ignoring the existing process to create a particular product it becomes feasible to develop the leanest, most performative process possible.

2.1.2 Product Based Design – Stages?

1. Scoping

- Select the business process to be redesigned
- Identify performance targets & limitations

2. Analysis

- Decompose information in the product specification into the product data model

3. Design

- Derive process designs matching design goals

4. Evaluation

- Verify & Validate product designs with end users
- Analyze estimated performance

2.1.3 Product Based Design – Stages?

Performance/importance matrix

(Martilla and James)

IMPORTANCE	High	Concentrate here	Maintain performance
	Low	Not important?	Possible overkill?
		Low	High
		PERFORMANCE	

TOPIC 03

Case Study

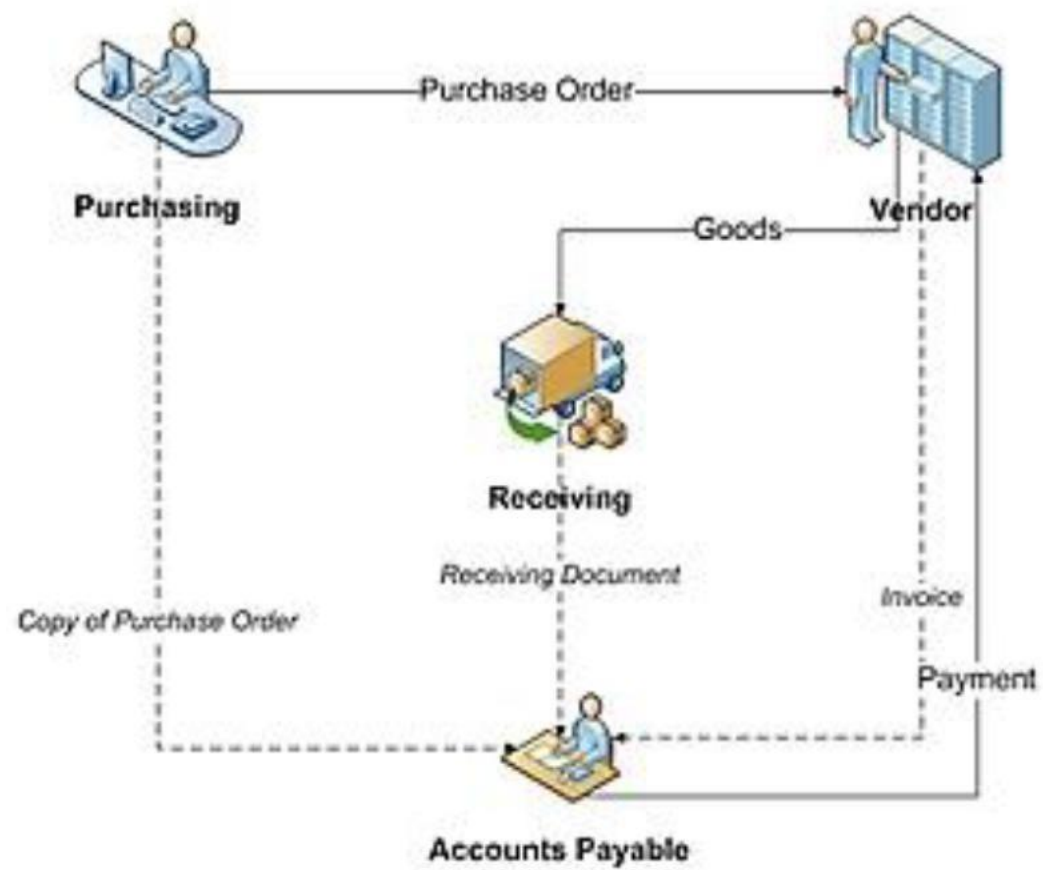


3.1 Ford Motors - Problem

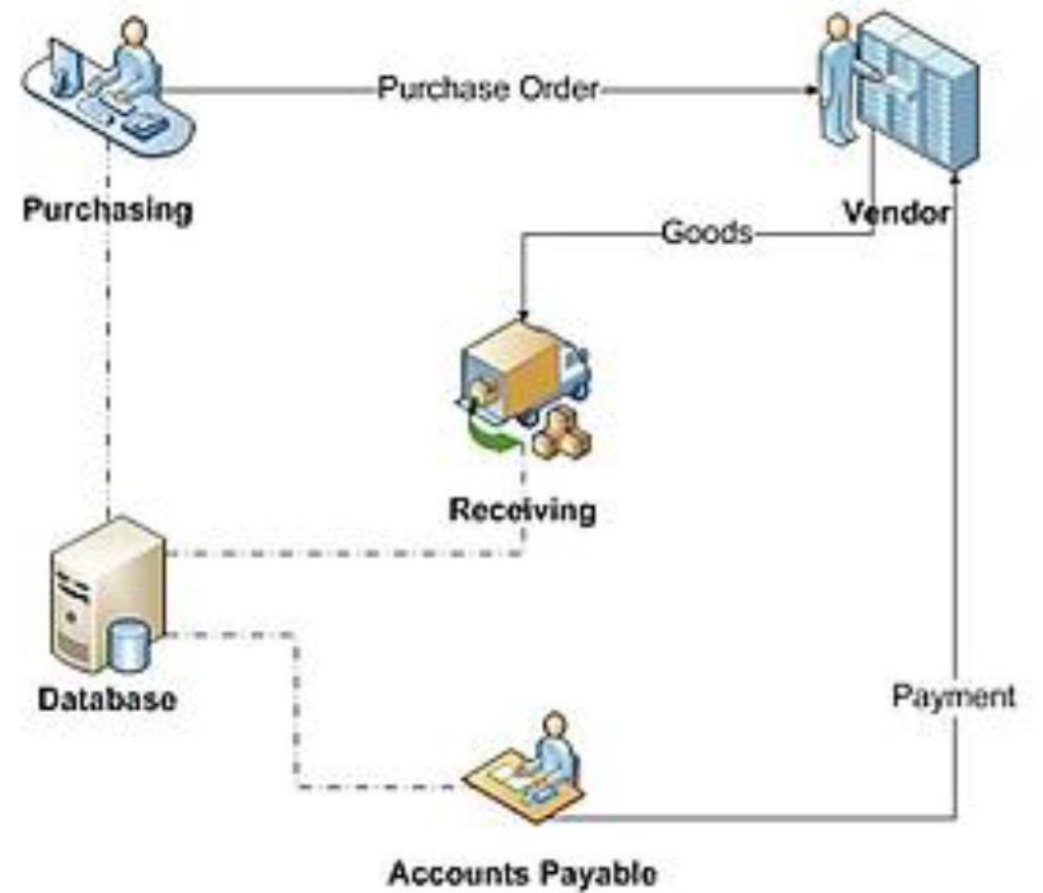
- Ford, an automobile manufacturing company.
- One of their findings was that the accounts payable department was not as efficient as it could be: their accounts payable division consisted of 500 people,
- Accordingly, Ford management set themselves a quantifiable goal: to reduce the number of clerks working in accounts payable by a couple of hundred employees.
- Then, they launched a business process reengineering initiative.
- When the purchasing department would write a purchase order, they sent a copy to accounts payable.
- Then, the material control would receive the goods, and send a copy of the related document to accounts payable.
- At the same time, the vendor would send a receipt for the goods to accounts payable.

3.2 Ford Motors - Solution

- With BPR, Ford completely recreated the process digitally.
 1. Purchasing issues an order and inputs it into an online database.
 2. Material control receives the goods and cross references with the database to make sure it matches an order.
 3. If there's a match, material control accepts the order on the computer.



Current Process



BPR Solution



IS 2206

LESSON 07

Recap...

Thank You!